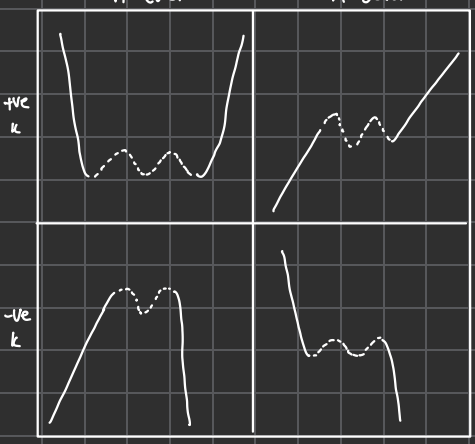


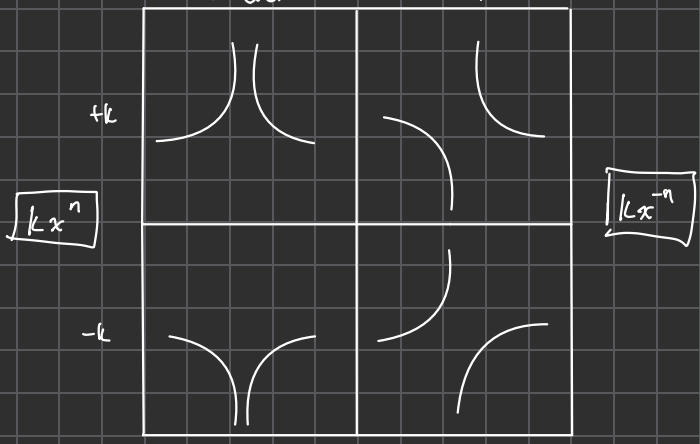
# 8.1: Shape of graphs.

Functional Notation:

$\Rightarrow y = x^2 \Leftrightarrow f(x) = x^2 \Leftrightarrow f: x \to x^2$




Function sketches can be checked on a calculator.




Direct Proportion: $y \propto x$ / $y = kx$

Indirect Proportion: $y \propto \frac{1}{x}$ / $y = \frac{k}{x}$

Other: $y \propto \frac{1}{x^2}$ / $y = \frac{k}{x^2}$

The equation of a transformed curve is made by combining effects of a transformation directly onto the equation of the curve.

Ex:

$y = x^2$

$T: \begin{pmatrix} 4 \\ 2 \end{pmatrix} \Rightarrow (x-4)^2 + 2$

$\Rightarrow \underline{x^2 - 8x + 18}$

## Graph Transformations

$f(x) \to f(x-t) \Rightarrow \begin{pmatrix} t \\ 0 \end{pmatrix}$

$f(x) \to f(x) + s \Rightarrow \begin{pmatrix} 0 \\ s \end{pmatrix}$

$f(x) \to f(x-t) + s \Rightarrow \begin{pmatrix} t \\ s \end{pmatrix}$

$f(x) \to -f(x) \Rightarrow$ Reflection in $y = 0$

$f(x) \to f(-x) \Rightarrow$ Reflection in $x = 0$

$f(x) \to af(x) \Rightarrow$ stretch, parallel to y, SF $a$.

$f(x) \to f(ax) \Rightarrow$ stretch, parallel to x, SF $\frac{1}{a}$

(Think of this as values will take larger y values, and hence the graph is stretched outwards)